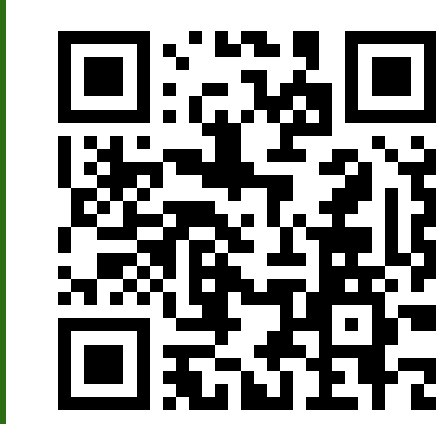




Background

The Goddard Earth Observing System Convection Allowing Model (GEOS-CAM) provides high resolution (~2 km) aerosol forecasts throughout the United States. While initially developed for hazardous weather prediction, GEOS-CAM features capabilities applicable to air quality and suborbital field campaigns. Despite this, the aerosol optical depth (AOD) in GEOS-CAM remains unevaluated against AOD observations.

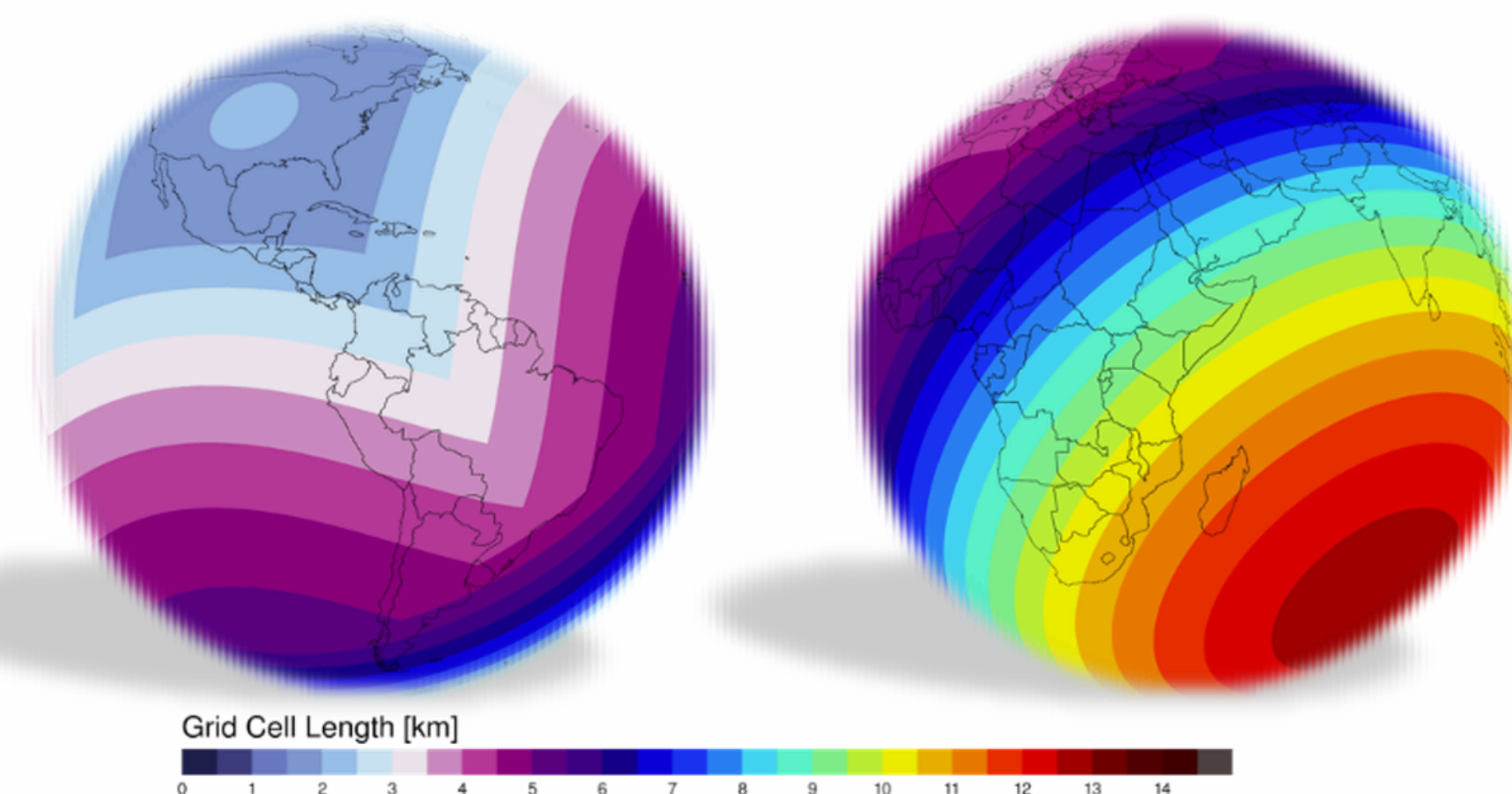


Fig 1: GEOS-CAM model resolution transitions from ~2 km over the United States to ~12.5 km as it moves away from the United States.

Goals & Objectives

- Create a Python utility for AeroApps to compute statistics between GEOS-CAM, GEOS-FP, and AERONET.
- Perform a basic evaluation of GEOS-CAM performance for aerosol prediction based on AERONET observations.
- Identify strengths and areas of improvement for GMAO to focus on.

Datasets and Methods

Datasets

- Aerosol Robotic Network (AERONET)
 - AERONET offers a worldwide network of stations with observations every 15 minutes.

Models	Spatial Resolution	Run Frequency	Aerosol Approach
GEOS-CAM	~2 km	Twice daily, 0Z and 12Z	Emissions and dynamical transport
GEOS-FP	~12 km	Twice daily, 0Z and 12Z	Assimilation, emissions and dynamical transport

Methods

- Gather GEOS-CAM and GEOS-FP AOD at times and locations with AERONET observations throughout the United States.
- Calculate RMSE, bias, and correlation at different times and forecast hours for each station.
- Identify modeled AOD evolution and compared with observations and analysis.

Results Across CONUS

Forecast Hours 36 - 48

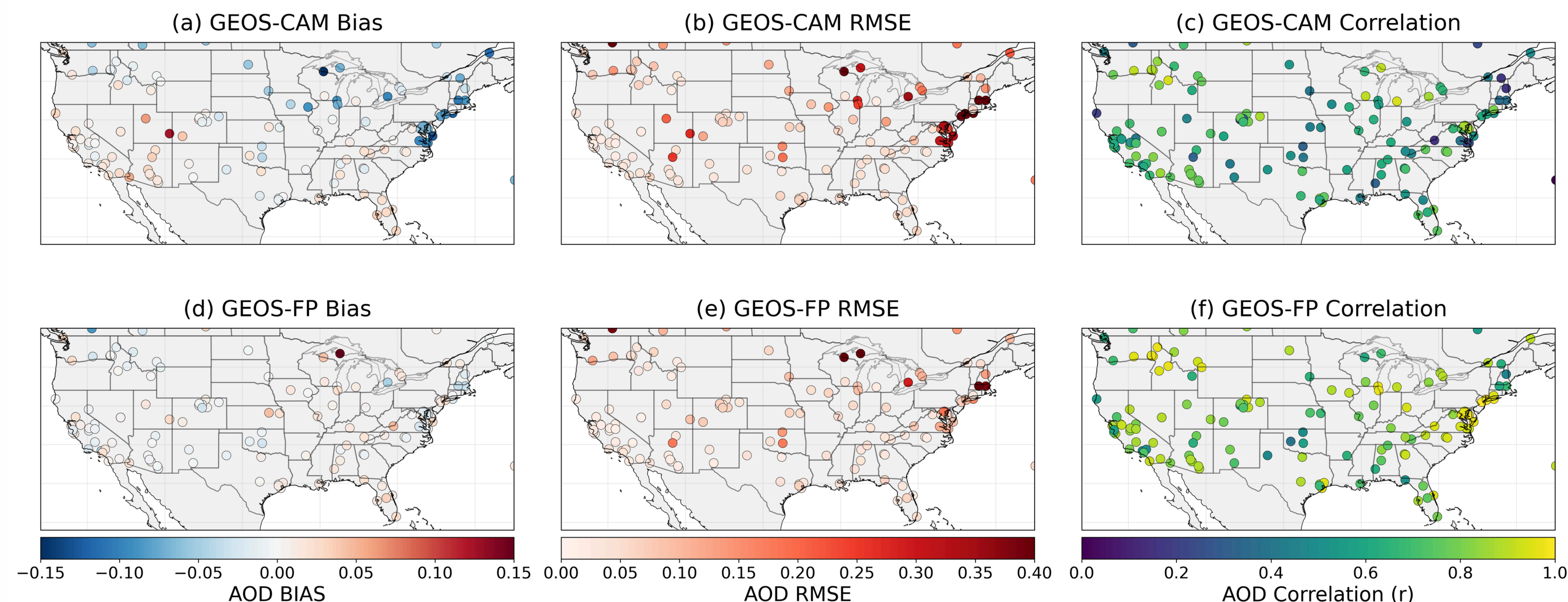


Fig 2: Evaluation of 550 nm AOD forecasts from GEOS-CAM (top) and GEOS-FP (bottom) compared to AERONET observations in the United States. The evaluation covers the 36- to 48-hour forecast period for 00z initializations from May 29th to July 27th and maps AOD Bias (a, d), Root Mean Square Error (RMSE) (b, e), and correlation (c, f).

Canadian Smoke Forecasts over GSFC

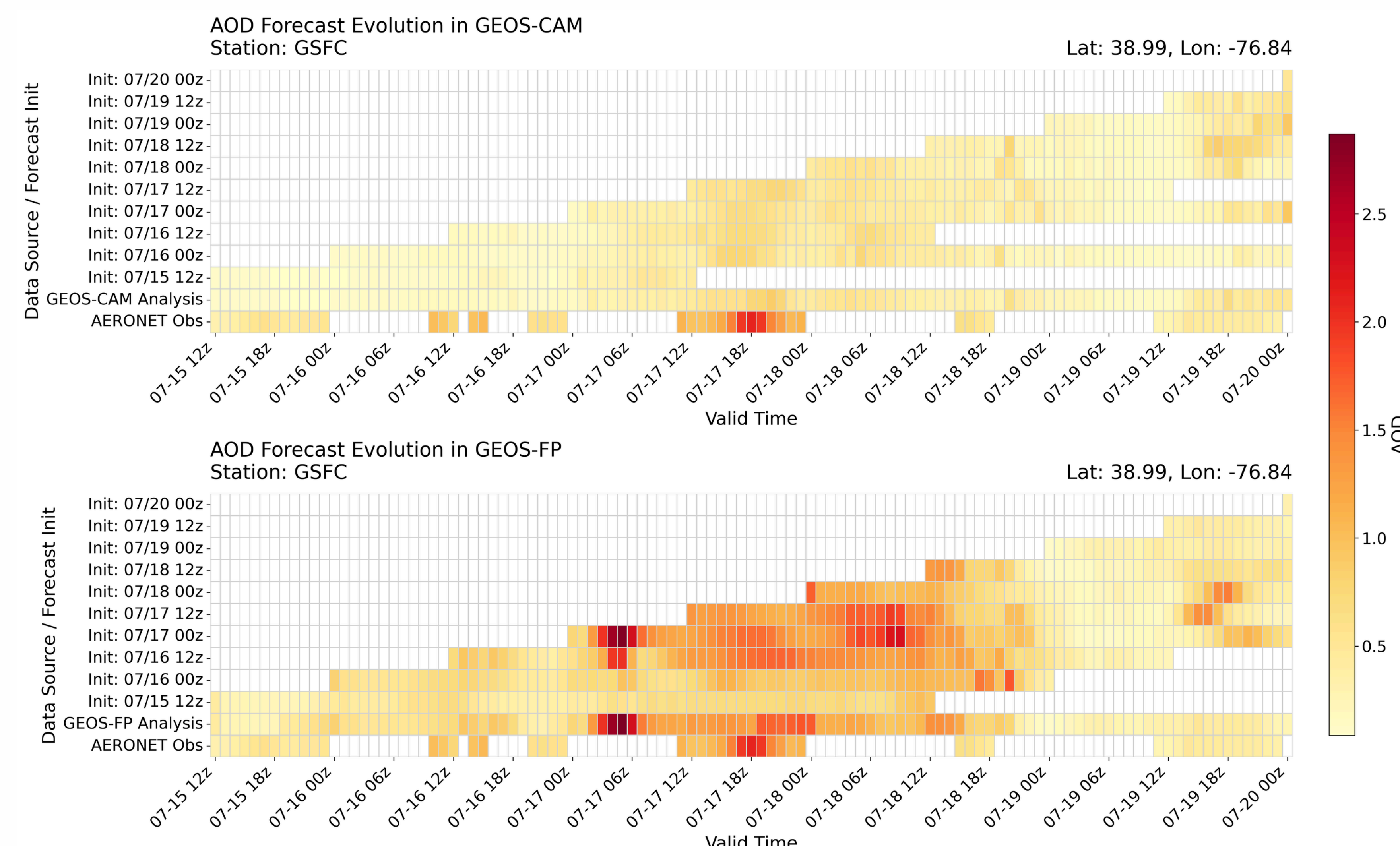


Fig 3: The evolution of AOD forecasts in GEOS-CAM (top) and GEOS-FP (bottom) from 12z on July 15th to 0z on July 20th at Goddard Space Flight Center. The bottom two rows of each model show model analysis and AERONET observations for comparison with the forecasts.

Key Takeaways

- GEOS-CAM aerosol forecasts lack AOD assimilation and is performing nearly as well as GEOS-FP aerosol forecasts.
- Bias and RMSE both show that GEOS-FP is performing slightly better than GEOS-CAM for aerosol forecasts.
- GEOS-CAM under estimates AOD in extreme smoke events.
- GEOS-FP predicts AOD spikes ~18 hours before the event.

Future Work

- Identify the influence of assimilation on forecast performance.
- Expand analysis to include fine particulate matter (PM_{2.5}).
- Evaluate GEOS-CAM aerosol composition.